

JESSICA MCFADYEN

AI RESEARCH SCIENTIST

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PROFILE

Cognitive neuroscientist and AI research scientist with experience spanning academia, health technology startups, and frontier AI evaluation. Expertise at the intersection of experimental science, computational modelling, product development, and large-scale real-world evaluation, with a focus on understanding complex intelligent systems—from human brains to modern AI models.

Track record of leading research programmes in mental health AI, developing and evaluating regulated AI products, and advancing methodologies for frontier AI evaluation. Experience translating foundational research into real-world impact across healthcare, technology, and policy-relevant domains.

ACHIEVEMENTS

AI Evaluation & Safety

- Leading fast (~3-month) research publication cycles on state-of-the-art AI evaluation science
- Ensuring scientific rigor in pre-deployment testing of frontier AI models from leading labs (OpenAI, Anthropic)
- Disseminating research findings to global AI security organisations and AI evaluation leaders

Research & Publications

- Landmark contributions in the fields of AI evaluation, computational psychiatry / neuroscience & AI-based precision medicine
- Proven track record in translational, real-world research (large-scale observational studies, experimental studies, and RCTs)
- Established commercial research collaborations with some of the largest healthcare systems in the world (NHS - UK, Kaiser Permanente - USA)

Leadership & Management

- Led clinical research & data insights team (7+ people) at Limbic
- General Manager of D2C product division at Limbic, leading cross-functional team of 10+ across UK and USA
- Head of Early Career Researcher Network, ARC Centre of Excellence at University of Queensland

EXPERIENCE

UK AI Security Institute

Senior Research Scientist
2026 - present

Limbic

Lead Clinical Researcher &
General Manager (D2C)
2024 - 2026

Clinical Evaluation Researcher
2024 - 2025

Alena

Senior Scientist
2022 - 2024

Max Planck UCL Centre for Computational Psychiatry and Ageing

Research Fellow
with Prof Ray Dolan
2019 - 2022

Queensland Brain Institute, University of Queensland

Postdoctoral Research Fellow
2018 - 2019

PhD
2015 - 2019

AWARDS

Dean's Commendation for Academic Excellence

PhD, University of Queensland
2016

\$105,000 (AUD) PhD Scholarship

University of Queensland
2015

Pushing Research Frontiers

- Applied novel neuroimaging and ML technology to reveal mechanisms (e.g., replay) of anxious decision-making (*Nature Neuroscience*)
- Utilised precision medicine approach to tailor treatment for social anxiety (Alena; *JMIR*)
- Deployed & refined clinical AI for CBT delivery (Limbic; *Nature Medicine*)
- Pioneered novel methods (DCM + dMRI) to understand neural pathways underlying psychiatric disorders (*eLife*, *Biological Psychiatry*)

Teaching, Mentorship, & Leadership

- Line management and mentorship of 6+ PhD students and early-career researchers in academic and industry settings.
- Designed long-term research strategy, directly reporting to CEO to guide company strategy (Alena, Limbic)
- Member of 8+ PhD milestone committees (UCL)
- Co-developed bachelor-level EdX course on cognitive biases (University of Queensland).

Translational & Applied Research

- Delivered clinical evidence for CE-marked medical devices (class IIa AI product, class I digital tool).
- Co-developed FDA breakthrough designation strategy for the first regulated AI healthcare intervention.
- Designed, built, and managed real-world AI healthcare products from concept → regulatory approval → market adoption.

Media & Outreach

- Presented research at conferences across 10+ countries
- Public opinion piece in *The Conversation* (2021)
- Spoke on *The Times* radio and Brisbane's 4ZZZ

Programming & Methods

- Fluent programmer with 15+ years experience in Python, SQL, MATLAB, Javascript, React, MATLAB, R, HTML, CSS, R, C#
- Quantitative modelling of behaviour and neuroimaging (DCM for fMRI/MEG/EEG, dMRI tractography, RL)
- AI product engineering & safety testing infrastructure
- Extensive web-development including online-experimental interfaces, web-design & real-time business analytics dashboards
- User research theory & practice and product design

Funding

- Key driver in a successful £7M Wellcome award (Limbic, 2025)
- Essential member of start-up fundraising meetings with potential venture-capitalist investors

PUBLICATIONS



18+ publications,
702+ citations

Selected Publications

J McFadyen, O Jorgensen, H Coppock, K Wei, C Ududec (2026) "How Inference Compute Shapes Frontier LLM Evaluation" *arXiv*

J McFadyen*, M Rollwage*, K Juechems, A Balogh, S Pisupati, TU Hauser, G Prichard, R Harper (in revision) "A Cognitive Layer Architecture Enables Patient-Facing (Mental) Health Support with LLMs" *Nature Medicine*

J McFadyen, Y Liu, RJ Dolan (2023) "Differential replay of reward and punishment paths predicts approach and avoidance" *Nature Neuroscience*

J McFadyen, RJ Dolan, MI Garrido (2020) "The influence of subcortical shortcuts on disordered sensory and cognitive processing" *Nature Reviews Neuroscience*

J McFadyen, RJ Dolan (2023) "Spatiotemporal precision of neuroimaging in psychiatry" *Biological Psychiatry*

J McFadyen*, MM Garvert*, S Linke, T McCloud, SS Meyer, S Sobanska (2025) "Safety and efficacy of modular digital psychotherapy for social anxiety: randomized controlled trial" *JMIR*

J McFadyen, J Habicht, LM Dina, R Harper, TU Hauser, M Rollwage (accepted) "AI-enabled conversational agent increases engagement with cognitive-behavioral therapy: A randomized controlled trial" *Communications Medicine*